

What is Research?

What are your ideas?

What is Research

- Google AI:
 - Research is the systematic, diligent investigation into a subject to discover new facts, verify existing information, or establish a deeper understanding. Derived from the Old French word *recerchier* (meaning to "search closely"), it encompasses both the process of gathering knowledge and the resulting findings.
- Purdue University
 - Research is the pursuit of new knowledge through the process of discovery. Scientific research involves diligent inquiry and systematic observation of phenomena. Most scientific research projects involve experimentation, often requiring testing the effect of changing conditions on the results. The conditions under which specific observations are made must be carefully controlled, and records must be meticulously maintained. This ensures that observations and results can be reproduced. Scientific research can be basic (fundamental) or applied.

Often broken down into two types

- Basic (National Science Foundation)
 - The objective of basic research is to gain more comprehensive knowledge or understanding of the subject under study, without specific applications in mind. In industry, basic research is defined as research that advances scientific knowledge but does not have specific immediate commercial objectives, although it may be in fields of present or potential commercial interest.
 - Almost all of our research will be of this type.

Often broken down into two types

- Applied (National Science Foundation)
 - Applied research is aimed at gaining knowledge or understanding to determine the means by which a specific, recognized need may be met. In industry, applied research includes investigations oriented to discovering new scientific knowledge that has specific commercial objectives with respect to products, processes, or services.

Can also be broken down by methods

- **Exploratory**
 - Used to investigate a problem or topic that isn't clearly defined. It helps gain background information and formulate hypotheses rather than providing final, conclusive answers.
 - Methods include qualitative research like interviews, focus groups, and case studies.

Can also be broken down by methods

- Descriptive
 - Designed to systematically observe and describe the characteristics of a population, phenomenon, or situation without manipulating variables.
 - Methods include surveys, observations, and cross-sectional studies.

Can also be broken down by methods

- Correlational
 - Examines the statistical relationship or association between two or more variables. It reveals patterns but cannot prove that one variable causes another.
 - Methods include statistical analysis of natural occurrences or archival data.

Can also be broken down by methods

- Experimental
 - A highly controlled scientific approach where researchers manipulate independent variables to observe their direct effect on a dependent variable. This is the only design that can establish true cause-and-effect relationships.
 - Methods include randomized controlled trials, detailed simulation, and laboratory testing.

How do you get started?

- Do some background reading on the subject.
 - “Library” is your friend.
 - There are videos on a lot of topics now!
- Talk to experts (mentors).
- See what others are doing in the field.
- Find a gap in our understanding or notice some interesting trend or ask a question that others haven't answered before.
- Devise a plan to fill the gap or to test the trend or to answer the question.

SIMCODES specific

- Using computational and theoretical methods to answer our questions
 - Simulations using physical models
 - Developing new models to be used – physical and/or AI based
 - Learning how to use computers to do what we want them to do
- Sharing our results with others
 - Repositories for software, data, models, and results
 - Creating a poster to discuss the research
 - Perhaps a paper

Research ethics

- Conflicts of interest
 - High ethical standards
 - Financial, family, security (for example)
 - If you have a conflict of interest, let us know
- Security
 - While most of what we do is open, you still need to ensure that you are keeping data secure and documented
- Integrity
 - Do not falsify work or data
 - Reproducibility and replicability
 - Acknowledge sources of information
 - No plagiarism – don't take credit for other's work
 - Acknowledge all people involved appropriately

Research ethics

- Treat everyone with respect
 - No harassment or discrimination
 - If issues, let your mentor or me and Ryan know

AI ethics

- Each mentor will have different expectations here, so have a discussion with them
- Be totally up front if you are using AI to generate code, data, text, ideas, etc.
 - While AI can accelerate research, it is still not ready to replace humans.
- Since this is a learning experience, you need to learn the content/approach before you use AI
 - Ex, learn basic Python before you have AI write code for you; otherwise you won't know when it is wrong or has "edge cases" that won't work
- AI is only as good as what is going in – garbage in/garbage out

AI ethics

- Test the results from AI
 - Does it pass the “does this make sense” test?
 - Think up tests that will challenge the results
- Learn how to give AI good prompts so that it won't hallucinate
- Be aware of biases in the models
- Question the results
- Interesting article here:
<https://link.springer.com/article/10.1007/s43681-024-00493-8>

Questions?

After lunch, around 1:30, you will be meeting with your mentors in either this room or Gilman 1761 (the Chemistry Help Center). The code for the latter is 8917. Write this down since you will need it!!